IGCSE 0607 Practice Questions — E5.4

Symmetry

Total Questions: 12 — **Total Marks:** 64 Marks

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1 Answer the following questions about the symmetry of two-dimensional shapes.

(a) Write down the number of lines of symmetry of a regular nonagon (a polygon with 9 sides).

..... **[1]**

(b) Write down the order of rotational symmetry of a parallelogram.

..... **[1]**

(c) Write down the mathematical name of a triangle that has exactly 1 line of symmetry and rotational symmetry of order 1.

..... **[1]**

2 Answer the following questions about 2D symmetry.

(a) Write down the order of rotational symmetry of a regular decagon.

..... **[1]**

(b) A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.
State the mathematical name of this quadrilateral.

..... **[1]**

(c) A shape has rotational symmetry of order 5.
Calculate the smallest angle of rotation that maps the shape onto itself.

..... ° **[1]**

3 Write down the symmetry properties for the following geometric shapes.

(a) A regular polygon has 15 sides.

Write down:

(i) the number of lines of symmetry,

..... **[1]**

(ii) the order of rotational symmetry.

..... **[1]**

(b) A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.

Write down the mathematical name of this quadrilateral.

..... **[1]**

(c) Write down the number of planes of symmetry of a solid right pyramid with a regular pentagonal base.

..... **[1]**

4 Consider the symmetry properties of the following 3D solid shapes.

(a) A right square-based pyramid has a base of side length 8 cm and perpendicular height 10 cm.
Write down the number of planes of symmetry for this pyramid.

..... **[1]**

(b) Write down the total number of planes of symmetry of a solid regular pentagonal prism.

..... **[2]**

(c) A right circular cone has base radius r and slant height l.
Describe fully its axis of symmetry.

..... **[2]**

5 The points $A(1, 6)$ and $B(7, 4)$ are two vertices of a quadrilateral $ABCD$.
The point $M(3, 2)$ is the centre of rotational symmetry of $ABCD$.
The quadrilateral $ABCD$ has rotational symmetry of order 4.

(a) Find the coordinates of C.

(..... , ...) **[2]**

(b) The line AC is a line of symmetry of the quadrilateral.

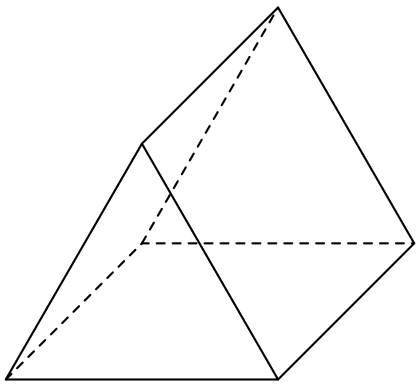
Find the equation of the line AC.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Calculate the exact area of the quadrilateral $ABCD$.

..... **[3]**

6 The diagram shows a solid right prism.
The cross-section of the prism is an equilateral triangle.



(a) State the number of planes of symmetry of this solid prism.

..... **[2]**

(b) State the order of rotational symmetry of the prism about the axis passing through the centroids of the two equilateral triangular faces.

..... **[1]**

(c) A different solid is a right regular hexagonal prism.

State the number of planes of symmetry of this hexagonal prism.

..... **[2]**

7 **(a)** A regular polygon has an exterior angle of 15°.

(i) Calculate the number of sides of this polygon.

..... **[2]**

(ii) Write down the number of lines of symmetry of this polygon.

..... **[1]**

(b) Write down the mathematical name of a quadrilateral that has exactly 1 line of symmetry and rotational symmetry of order 1.

..... **[1]**

(c) A solid cone has a base radius of 5 cm and a perpendicular height of 12 cm.

State the number of planes of symmetry of this cone.

..... **[1]**

8 **(a)** A solid wooden block is in the shape of a right prism.
The cross-section of the prism is a regular octagon.

(i) Write down the order of rotational symmetry of the cross-section.

..... **[1]**

(ii) State the total number of planes of symmetry of this right prism.

..... **[2]**

(b) A different solid is a right pyramid with a regular polygon of n sides as its base.
The pyramid has exactly 12 planes of symmetry.

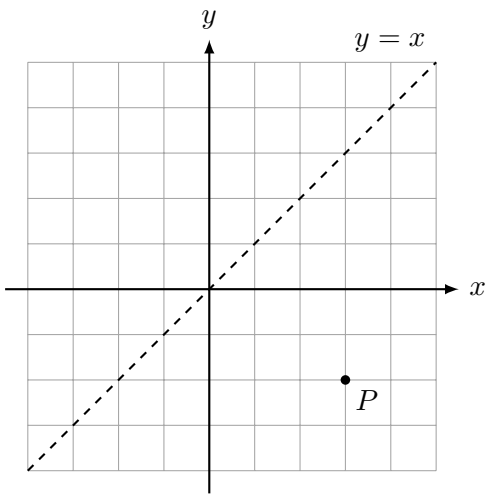
(i) Write down the value of n.

$n =$ ..... **[1]**

(ii) Calculate the sum of the interior angles of the base of this pyramid.

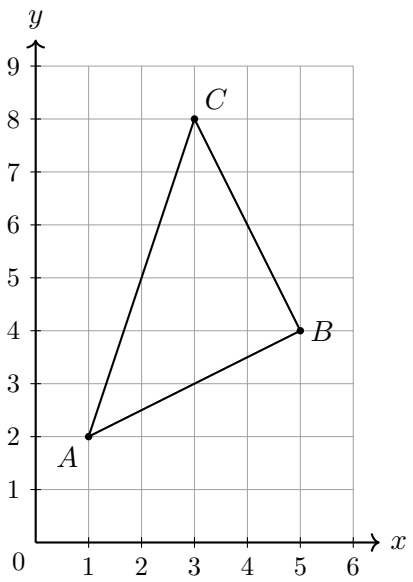
..... ° **[2]**

(c) A shape S is drawn on a coordinate grid.
The shape S has a line of symmetry with equation $y = x$.
One vertex of the shape is at the point $P(3, -2)$.
Write down the coordinates of the image of P after a reflection in this line of symmetry.



(..... , ...) **[2]**

9 Triangle ABC has vertices at $A(1, 2)$, $B(5, 4)$ and $C(3, 8)$.



(a) Show that triangle ABC is an isosceles triangle.

[3]

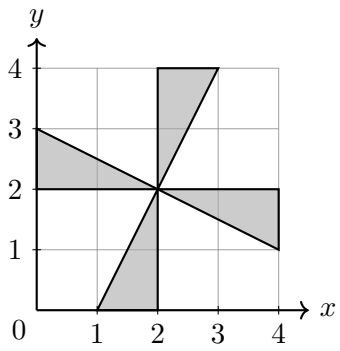
(b) Triangle ABC has exactly one line of symmetry.
Find the equation of this line of symmetry.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(c) A different shape is a regular polygon with n sides.
It has exactly 15 lines of symmetry.
Calculate the size of one interior angle of this polygon.

..... ° **[2]**

10 The diagram shows a shaded shape on a coordinate grid.



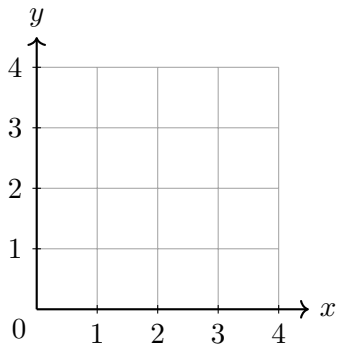
(a) Write down the order of rotational symmetry of the shaded shape.

..... **[1]**

(b) Write down the number of lines of symmetry of the shaded shape.

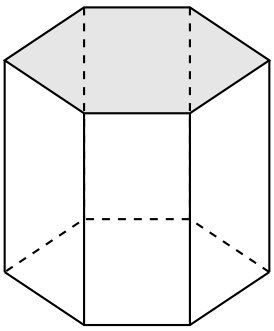
..... **[1]**

(c) On the grid below, draw the image of the shape after a reflection in the line $x = 2$.



[2]

11 The diagram shows a solid regular hexagonal prism.



(a) Write down the number of planes of symmetry of this regular hexagonal prism.

..... **[2]**

(b) Write down the order of rotational symmetry of the prism about the vertical axis passing through the centres of its hexagonal faces.

..... **[1]**

(c) A different solid is a right square-based pyramid.

Write down the number of planes of symmetry of this pyramid.

..... **[1]**

12 A rhombus $ABCD$ has its centre at the origin $(0, 0)$.

(a) The line of symmetry passing through vertices A and C has the equation $y = 3x$.
Find the equation of the line of symmetry passing through vertices B and D.

$y =$ ..... **[2]**

(b) Write down the order of rotational symmetry of the rhombus $ABCD$.

..... **[1]**

(c) A different shape is a regular polygon with n sides.
The number of lines of symmetry of this polygon is equal to n.
The interior angle of this polygon is $165°$.

Calculate the value of n.

$n =$ ..... **[3]**